

InventoriSystem

System Documentation

A Complete Inventory Management System

Build Duration: May 5-9, 2026 (5 Days)

Technologies: HTML5, CSS3, JavaScript (Vanilla)

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1. Project Overview

InventoriSystem is a web-based inventory management system designed to help companies track and manage their products, monitor stock levels, generate reports, and configure system settings.

The system was built from scratch using vanilla HTML, CSS, and JavaScript without any external frameworks or libraries, making it lightweight, fast, and easy to deploy.

1.1 Project Objectives

- Create a fully functional inventory management frontend
- Implement responsive design for desktop and mobile devices
- Provide a clean and intuitive user interface
- Support light and dark theme modes
- Enable easy deployment via GitHub Pages

2. System Requirements

2.1 Browser Requirements

- Google Chrome (latest version)
- Mozilla Firefox (latest version)
- Safari (latest version)
- Microsoft Edge (latest version)

2.2 Device Requirements

- Desktop computers (1024px+ screen width)
- Tablet devices (768px - 1024px screen width)
- Mobile phones (below 768px screen width)

2.3 Technical Requirements

- No server-side processing required (frontend-only)
- SessionStorage for login state management
- LocalStorage for theme preference persistence
- No database connection needed (mock data in JavaScript)

3. Architecture & Design

3.1 System Architecture

The system follows a multi-page application (MPA) architecture where each page is a separate HTML file sharing common CSS and JavaScript resources.

3.2 Design Principles

- Modular code organization with separate files for each concern
- Shared CSS and JavaScript across all pages for consistency
- Component-based styling using CSS custom properties (variables)
- Progressive enhancement approach for browser compatibility
- Mobile-first responsive design methodology

3.3 Color Scheme

Light Theme:

- Primary Orange: #FF6600
- Background: #FFFFFF
- Sidebar Background: #1A1A2E (dark blue)
- Card Background: #FFFFFF
- Text Primary: #333333

Dark Theme:

- Primary Orange: #FF6600
- Background: #1A1A2E (dark blue)
- Sidebar Background: #0F0F1A (darker blue)
- Card Background: #16213E
- Text Primary: #FFFFFF

4. File Structure

The project follows a clean and organized folder structure:

4.1 Directory Layout

```
InventoriSystem/  
- index.html (redirect to login)  
- pages/  
  - login.html  
  - dashboard.html  
  - products.html  
  - reports.html  
  - settings.html  
- assets/  
  - css/style.css  
  - js/script.js  
  - images/LogoCompany.jpg  
- plans/2026-05-07-inventory-system-design.md
```

4.2 File Responsibilities

style.css: Contains all styling, themes, animations, and responsive rules

script.js: Handles all JavaScript functionality, navigation, and data management

Each HTML file: Represents a single page of the application

5. Technology Stack

5.1 Core Technologies

HTML5 (HyperText Markup Language)

- Semantic HTML elements (header, nav, main, section, footer)
- Form validation attributes
- Meta viewport for responsive design
- SVG icons embedded inline

CSS3 (Cascading Style Sheets)

- CSS Custom Properties (variables) for theming
- Flexbox and CSS Grid for layouts
- CSS Animations and Transitions
- Media queries for responsive design
- Pseudo-elements and pseudo-classes for interactivity

JavaScript (ES6+)

- DOM manipulation and event handling
- sessionStorage and localStorage for data persistence
- Template literals for dynamic HTML generation
- Arrow functions and destructuring
- Array methods (map, filter, reduce)

5.2 No External Dependencies

The system was built entirely without external libraries, frameworks, or CDN resources, ensuring:

- Zero external requests for faster loading
- No dependency on third-party servers
- Full control over code and styling
- Easy to understand and modify

6. Day-by-Day Implementation

6.1 Day 1 - May 5, 2026 (Requirements & Planning)

- Analyzed user requirements and project scope
- Clarified design expectations with user via questions
- Created design specification document (plans/2026-05-07-inventory-system-design.md)
- Designed file structure layout
- Established color scheme and theme strategy

6.2 Day 2 - May 6, 2026 (Core Infrastructure)

- Created folder structure (assets/css, assets/js, assets/images, pages)
- Created shared CSS stylesheet (style.css) with theme variables
- Created shared JavaScript file (script.js) with core functions
- Implemented CSS custom properties for theming
- Built basic responsive layout system

6.3 Day 3 - May 7, 2026 (Page Development)

- Created login.html with centered card design
- Created dashboard.html with stats cards and recent products table
- Created products.html with CRUD interface and modal forms
- Created reports.html with placeholder charts and tabs
- Created settings.html with profile, company info, and toggles
- Copied LogoCompany.jpg to assets/images folder

6.4 Day 4 - May 8, 2026 (Enhancements & Animations)

- Updated sidebar styling from orange to dark blue (#1A1A2E)
- Added hover effects on buttons with ripple animation
- Added card hover animations with gradient top border
- Added table row slide animation on hover
- Added modal slide-up and scale entrance effects
- Added floating logo animation on login page
- Added navigation highlight sweep animation
- Added status text underline animation on hover
- Added scrollbar orange highlight on hover

6.5 Day 5 - May 9, 2026 (Deployment & Documentation)

- Initialized Git repository in project folder
- Set up remote origin to github.com/NightmareCode/Airgas
- Pushed all files to GitHub repository
- Configured GitHub Pages for deployment
- Created this documentation PDF
- Verified website is accessible at NightmareCode.github.io/Airgas

7. Features & Functionalities

7.1 Navigation System

- Fixed sidebar navigation on desktop
- Collapsible sidebar with hamburger menu on mobile
- Active state highlighting for current page (orange background)
- Smooth transitions between pages
- Overlay backdrop when sidebar is open on mobile
- Orange highlight sweep animation on hover

7.2 Theme System

- Light theme: White background with dark blue sidebar
- Dark theme: Dark blue background with orange accents
- Theme preference saved to LocalStorage
- Instant theme switching without page reload
- Theme toggle accessible from all pages

7.3 Animations & Effects

- Button ripple effect on click
- Card hover lift and shadow effects
- Table row slide animation on hover
- Modal slide-up and scale entrance
- Logo floating animation on login page
- Background pulse animation on login page
- Navigation highlight sweep animation
- Status text underline animation on hover
- Scrollbar orange highlight on hover

7.4 Form Handling

- Input focus states with color transitions
- Form validation feedback (shake animation on error)
- Hover effects on form elements
- Clear form on modal close
- Pre-populated form for edit operations

8. User Interface Design

8.1 Layout Structure

- Sidebar: Fixed 250px width on desktop, collapsible on mobile
- Main Content: Fluid width with consistent padding
- Mobile Header: Fixed top bar with hamburger menu
- Overlay: Semi-transparent backdrop for mobile sidebar

8.2 Component Design

Cards:

- Rounded corners (12px radius)
- Subtle shadow for depth
- Top border gradient animation on hover
- Consistent padding (24px)

Buttons:

- Rounded corners (8px radius)
- Orange primary color with hover darkening
- Scale and shadow lift on hover
- Ripple effect on click

Tables:

- Alternating row backgrounds
- Row hover highlighting
- Horizontal scroll on mobile
- Sticky header on scroll

8.3 Typography

- Font Family: Segoe UI, Roboto, Helvetica Neue, Arial (system fonts)
- Heading sizes: 2rem (h1) to 1rem (h6)
- Body text: 16px base size with 1.6 line height
- Font weights: 400 (normal), 500 (medium), 600 (semibold), 700 (bold)

9. Responsive Design

9.1 Breakpoints

- Desktop: Above 1024px - Full sidebar, 4-column stats grid
- Tablet: 768px - 1024px - Collapsible sidebar, 2-column stats grid
- Mobile: Below 768px - Hidden sidebar with overlay, single column layout
- Small Mobile: Below 480px - Compact stat cards, centered content

9.2 Mobile Adaptations

- Hamburger menu replaces fixed sidebar
- Sidebar slides in from left with overlay backdrop
- Stats cards stack vertically
- Tables enable horizontal scrolling
- Form inputs expand to full width
- Report tabs stack vertically

9.3 Touch Considerations

- Minimum touch target size of 44px for interactive elements
- Adequate spacing between clickable items
- Hover states also trigger on touch for compatibility

10. Deployment

10.1 GitHub Pages Deployment

The system is deployed using GitHub Pages, a static site hosting service:

- Repository: `github.com/NightmareCode/Airgas`
- Branch: `master` (configured for GitHub Pages)
- Entry point: `index.html` (redirects to `pages/login.html`)
- URL: `https://NightmareCode.github.io/Airgas`

10.2 Deployment Process (May 9, 2026)

- Initialized Git repository with `'git init'`
- Added remote origin with GitHub token authentication
- Staged all files with `'git add -A'`
- Committed with initial message: `'Initial commit - Complete inventory management system'`
- Pushed to master branch with `'git push -u origin master --force'`
- GitHub Pages automatically detected and published the site

10.3 Future Backend Integration

The system is designed to be easily extended with backend functionality:

- Replace mock data arrays with API calls
- Implement actual authentication with server validation
- Connect to database for persistent storage
- Add server-side form validation
- Implement real-time updates with WebSockets

End of Documentation